

Physics-Aware Hierarchical 3DGS Compression: An Efficient Representation for Aerospace Objects

Dazhuang Yang^{1b}, Tao Zhao^{1b}, Yizhai Zhang^{1b}, Ganghui Shen^{1b}, *Member, IEEE*, Guotao Fang^{1b},
Panfeng Huang^{1b}, *Senior Member, IEEE*, Binglu Wang^{1b} *Member, IEEE*

Abstract—While 3D Gaussian Splatting (3DGS) has revolutionized real-time rendering with high visual fidelity, its unstructured and discrete nature poses significant challenges for efficient storage and transmission, especially in bandwidth-constrained space-ground communications. Existing compression paradigms rely on signal redundancy, often leading to geometric collapse on complex structures at extreme bitrates. To overcome these limitations, we propose a physics-aware hierarchical 3DGS compression framework tailored for aerospace objects. Our approach reconfigures the compression paradigm, departing from conventional pixel-level entropy coding toward semantic-oriented component reorganization. Specifically, a geometry layer condenses repetitive structures into compact meta-templates via a coarse-to-fine cascade: spectral divergence pruning, robust rigid alignment, and alternating differentiable refinement. Subsequently, an appearance layer exploits material sparsity through component-wise manifold vector quantization. Extensive evaluations on the Spacecraft Pose Estimation and 3D Reconstruction Dataset (SPARD) demonstrate that our method effectively addresses the rate-distortion bottlenecks of current compression paradigms. It achieves an approximately $8\times$ storage compression ratio, compressing the explicit 3D representation to about 2 MB while maintaining high-fidelity visual quality above 30 dB. By synergistically decoupling spatial topology and material statistics, our method preserves structural integrity and delivers a robust, high efficiency 3D representation for on-orbit servicing tasks.

Index Terms—Physics-Aware, Model Compression, Hierarchical, Aerospace Objects, 3DGS.

I. INTRODUCTION

THE deployment of mega-constellations and the increasing amount of space debris have heightened interest in on-orbit servicing missions. Whether for anomaly detection, collision avoidance, or dexterous manipulation, servicing satellites must perceive the geometric structure of target objects in a real-time and intuitive manner [1]. However, the limited onboard computational resources and communication bandwidth of servicing satellites pose significant challenges to the

efficient representation and transmission of high-fidelity 3D models.

One straightforward solution is to offload computational tasks to ground stations due to limited onboard computational resources. As illustrated in Fig. 1, within the integrated space-ground computing architecture, multi-view images captured by the servicing satellite are downlinked to ground receiving stations. Then, high-performance servers perform resource-heavy 3D reconstruction and optimization. The compressed model is uploaded back to the satellite, enabling efficient storage, rendering, and downstream tasks under limited onboard resources.

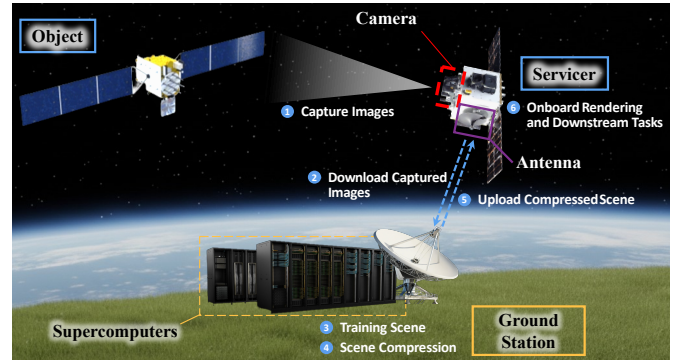


Fig. 1: Integrated space-ground computational architecture for on-orbit servicing missions. A servicing satellite captures multi-view images of a target. Large-scale computation is performed on ground supercomputers.

Computationally intensive methods, such as deep learning and high-fidelity 3D reconstruction, become feasible under the emerging paradigm. While implicit neural representations, *e.g.* Neural Radiance Fields (NeRF) [2], have set the benchmark for visual quality, their low inference speeds render them unsuitable for tasks requiring immediate visual feedback. To address this inference bottleneck, 3D Gaussian Splatting (3DGS) [3] has recently emerged as a promising alternative. By explicitly representing the scene with 3D Gaussian primitives and leveraging a highly parallelizable rasterizer, 3DGS achieves real-time rendering speeds while maintaining high visual fidelity, making it ideal for time-sensitive on-orbit servicing tasks.

However, 3DGS fundamentally relies on an unstructured, discrete representation. This lack of structure results in massive storage costs, making raw 3DGS transmission impractical

*This work was supported by the National Natural Science Foundation of China under Award 62573351, 62022067, 62403383. (Corresponding author: Yizhai Zhang.)

Dazhuang Yang is with Shaanxi Province Innovation Team of Intelligent Robotic Technology, School of Automation, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, China (e-mail: yang_dazhuang@mail.nwpu.edu.cn)

Tao Zhao, Yizhai Zhang, Ganghui Shen, Guotao Fang, Panfeng Huang, and Binglu Wang are with Shaanxi Province Innovation Team of Intelligent Robotic Technology, School of Astronautics, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, China (e-mail: t_zhao_ai@163.com; zhangyizhai@nwpu.edu.cn; shenganghui@nwpu.edu.cn; gtfang@nwpu.edu.cn; pfhuang@nwpu.edu.cn; wbl921129@gmail.com)

for bandwidth-constrained space links. Existing 3DGS compression techniques rely on statistical redundancy and ignore the intrinsic structure of the target [4,5]. At extreme compression ratios, such methods often lead to severe geometric collapse and visual artifacts, which are unacceptable for safety-critical space applications.

To overcome these limitations, this work explores a paradigm shift from signal compression to semantic reassembly. Unlike natural scenes, aerospace objects are highly structured industrial products. Such products naturally exhibit two strong physical priors: a finite library of surface materials and highly repetitive rigid components. These physical properties offer a unique opportunity for extreme data reduction.

Guided by this insight, we propose a physics-aware hierarchical 3DGS compression framework tailored for aerospace objects. As illustrated in Fig. 2, this framework maps the unstructured Gaussian field into a low-dimensional semantic manifold. We explicitly decouple the representation into a geometry layer and an appearance layer, each designed to exploit the underlying physical priors. The geometry layer exploits rigid kinematic constraints to replace repetitive structures with shared templates. Simultaneously, the appearance layer leverages vector quantization to map continuous appearance attributes onto a compact material codebook. To ensure high-fidelity reconstruction, a multi-domain consistency optimization is employed, enforcing alignment across physical and visual domains.

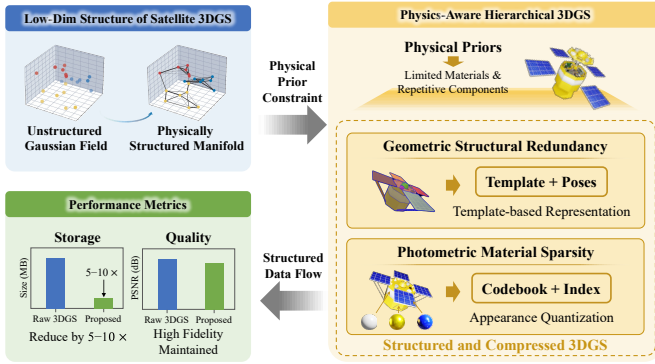


Fig. 2: Representation philosophy of the proposed physics-aware hierarchical 3DGS framework. By imposing industrial physical priors on unstructured Gaussian field, the scene is decoupled into a compact and structured representation.

The main contributions of this study are summarized as:

- A physics-aware hierarchical compression framework for 3DGS, specifically designed for stringent computational and telemetry constraints in on-orbit perception scenarios.
- A structural decoupling mechanism that maps unstructured Gaussian fields into compact, physically structured geometry and appearance manifolds.
- Extensive experimental validation demonstrates that the proposed representation can achieve high compression ratios without compromising reconstruction quality.

II. RELATED WORK

In this section, we first discuss the evolution from implicit to explicit 3D representations, highlighting the advantages of 3DGS. Then, current compression techniques are analyzed, emphasizing their limitations in aerospace applications. Finally, we examine physics-aware approaches to reveal the disparity between semantic understanding and compression efficiency, motivating our proposed physics-aware hierarchical framework.

A. Efficient 3D Representations for Aerospace Objects

The trade-off between rendering quality, inference speed, and storage efficiency is the central challenge for space-ground integrated perception.

Historically, traditional point clouds have served as the dominant paradigm for spatial target perception [6]. However, these traditional discrete representations fundamentally fail to model continuous surfaces and complex specular reflections inherent in real-world scenarios [7,8].

Implicit neural representations, represented by NeRF [2], model scenes as continuous volumetric fields parameterized by neural networks. Due to high visual fidelity, some researches attempt to apply NeRF to aerospace targets. Liu *et al.* [9] incorporates adaptive background suppression and hybrid encoding to achieve high-fidelity novel view synthesis of satellite targets. Han *et al.* [10] and Yang *et al.* [11] simultaneously estimate 6-DoF poses and reconstruct non-cooperative spacecraft with NeRF-based representations. Although implicit neural representations achieve high visual fidelity with compact storage, their low inference speeds limit real-time applications [12]. A series of works has attempted to accelerate NeRF rendering through hash encoding [13], hybrid representations [14], and sampling reduction [15]. However, these methods still struggle to meet the stringent real-time requirements of on-orbit servicing tasks.

As a disruptive advancement, 3DGS [3] revisits explicit representations. By modeling scenes with 3D Gaussians and leveraging tile-based rasterization, 3DGS achieves real-time rendering (> 100 FPS) with high visual quality. This breakthrough has recently been adapted for space applications. For instance, Li *et al.* [16] integrates 2DGS with Bayesian view estimation and orbital mechanics to achieve active 3D reconstruction of space targets. Satellite-GS [17] enhances the Gaussian representation with physical deblurring and detail super-resolution modules to combat severe space-to-space optical degradation. However, as mentioned in Section I, this performance comes at the cost of an unstructured, discrete representation requiring millions of primitives [18]. This characteristic makes raw 3DGS impractical to deploy directly on servicer terminals and inefficient to transmit over bandwidth-limited space links [19]. Therefore, efficient compression techniques are essential to unlock the full potential of 3DGS in aerospace applications.

B. Compression of 3DGS Representations

Early research on 3DGS compression was primarily driven by the reduction of numerical redundancy. This strategy com-

bines gradient-guided densification and opacity-based pruning to remove low-contributing Gaussians. LightGS [20] and MiniGS [21] identify and discard redundant Gaussians based on opacity thresholds, gradient magnitudes, or global importance scores projected onto camera views. While effective in reducing data volume, this aggressive elimination is often detrimental to structural integrity [22].

Vector Quantization (VQ) maps continuous attribute vectors to a discrete set of codewords stored in a codebook [5,23]. This strategy is particularly effective for compressing appearance attributes, which often exhibit significant redundancy across a scene [20,24]. Notable works like CompGS [23] and Niedermayr *et al.* [24] employ K-means clustering to construct local codebooks for Spherical Harmonics (SH) and colors. LightGS [20] further optimizes VQ efficiency by simplifying codebook updates and introducing adaptive codeword allocation, lowering training overhead. EAGLES [25] combines VQ with pruning to achieve $10\times$ storage reduction. However, these methods fundamentally operate on the entangled Gaussian field, constructing codebooks based on statistical similarity rather than intrinsic material semantics. Consequently, they fail to leverage the global material consistency unique to industrial targets, motivating our adoption of quantization within a structured physics-aware framework.

Research attention has also expanded toward structured models that exploit spatial topology to further enhance compression ratios. The anchor-based architecture, represented by Scaffold-GS [26] and HAC series [27,28], has emerged as a dominant paradigm in this direction. In this framework, the scene is represented by a sparse set of anchor points that are optimized to capture the underlying structure [29]. While Scaffold-GS [26] reduces the number of stored entities, it does not inherently perform entropy-based compression. The anchors and their neural network weights are still typically stored as 32-bit floats. Furthermore, HAC [27] introduces entropy modeling to the anchor level. It employs a hash grid that can be queried at any anchor location to obtain an interpolated hash feature. HAC++ [28] improves upon HAC by introducing channel-wise autoregressive modeling and an adaptive masking strategy that eliminates invalid Gaussians and anchors during training. Additionally, SHTC [29] explores transform coding techniques, mapping attributes from the spatial to decorrelated frequency domain. ContextGS [30] introduces a hierarchical encoding, using coarse-level anchor points to predict the distribution of fine-level anchor points.

Fundamentally, approaches based on pruning, quantization, or anchoring are formulated around local statistical regularities. By neglecting object semantics, these approaches lead to geometric collapse at extreme compression ratios, violating the strict structural integrity required for space missions [31]. This limitation motivates a shift from signal-level compression toward a physics-aware semantic reassembly framework, in which geometry and appearance are explicitly decoupled and organized hierarchically according to industrial structural priors.

C. Structure-Aware and Compositional 3DGS

To mitigate the artifacts arising from the unstructured nature of 3DGS, geometric regularization is increasingly adopted to enforce surface integrity. 2DGS [32] replaces volumetric ellipsoids with oriented planar surfels to improve geometric accuracy, while SuGaR [33] aligns Gaussians with explicit meshes to facilitate physics simulation. For man-made environments, PlanarGS [34] and MG-SLAM [35] incorporate planar or Manhattan World priors to rectify geometry in textureless regions. However, the scene is typically reconstructed as a monolithic surface by these methods. The part-level repetition inherent in industrial assemblies is not explicitly modeled. Consequently, the kinematic redundancy of repetitive rigid components remains unexploited, limiting the potential for extreme compression.

Beyond geometric regularization, semantics has been integrated into the Gaussian field to enable object-level perception. Feature 3DGS [36] established the paradigm of embedding high-dimensional vectors directly onto Gaussian primitives. This mechanism demonstrated that 3DGS can distill semantic knowledge via differentiable rasterization. LangSplat [37] compresses semantic features into a compact latent space and leverages SAM-derived region priors for improved boundary coherence, at the cost of additional decoding latency and persistent view-dependent inconsistencies. SAGA [38] further improves efficiency through low-dimensional affinity representations and a scale-gated mechanism across different granularities, thus achieving near real-time performance. Despite these advances, all three methods remain signal-driven. Semantic features are treated as auxiliary attributes attached to Gaussian primitives, without exploiting structural regularities inherent in repetitive rigid components.

Conceptually, the instantiation of standardized structural templates is isomorphic to the problem of pose estimation. Template matching remains a dominant paradigm within the 3D Gaussian Splatting framework for resolving such rigid alignment tasks. GS-Pose [39] constructs a library of Gaussian templates from a small set of posed reference views, and adopts rotation-aware embedding vectors as retrieval templates for rough 6D pose estimation. Specifically, GS-Pose employs a commonly used differentiable optimization strategy for comparing rendering results [40], which is essentially a template-based paradigm [11]. Departing from iterative refinement, 6DGS [41] constructs attention-guided 2D–3D correspondences between image features and Gaussian-generated ray bundles, enabling closed-form camera pose recovery without explicit initialization. These methods match Gaussian templates to observed data, focusing on pose estimation, but do not exploit the redundant structures of aerospace objects. In contrast to prior template-based formulations, our framework instead introduces unsupervised discovery of internal geometric templates to enable physics-aware compression.

III. PROPOSED METHOD

This section details the proposed physics-aware hierarchical 3DGS compression framework. Building upon the fundamentals of 3DGS, the framework is conceptualized through a systematic pipeline: an initial semantic-spatial decomposition of

the unstructured Gaussian field, followed by the construction of a decoupled hierarchical representation, and concluded with the corresponding optimization strategies to enforce physical constraints.

A. System Overview

Our framework operates within an asymmetric architecture consisting of a ground station and a servicing satellite. The ground station is responsible for constructing the compact 3DGS representation from raw sensor data, while the servicing satellite utilizes this representation for real-time rendering and navigation tasks.

Given an unstructured Gaussian field $\mathcal{G}$, our goal is to learn a disentangled and structured representation $\hat{\mathcal{G}} = \{\mathcal{G}_{geo}, \mathcal{G}_{app}\}$ to minimize the storage cost $R(\cdot)$ while maximizing the visual quality $Q(\cdot)$ under physics constraints:

$$\max_{\hat{\mathcal{G}} \in \Omega_p} \left(Q(\hat{\mathcal{G}}) - \gamma_r R(\hat{\mathcal{G}}) \right) \quad (1)$$

where Ω_p denotes the feasible set defined by rigid kinematic and material sparsity. γ_r is a hyperparameter that controls the trade-off between quality and compression.

To solve this non-convex optimization problem, we propose a framework to progressively transform the unstructured description of Gaussian fields into a compact and structured representation, as illustrated in Fig. 3.

Stage 1: Pre-processing. Starting with the reconstructed 3D Gaussian field, semantic and spatial decomposition is conducted to identify functional components. Specifically, a decoupled optimization paradigm is employed: we first establish a high-fidelity geometric foundation via photometric supervision, and then distill 2D semantic priors into fixed primitives for decomposition.

Stage 2: Hierarchical Compression. Building upon the decomposed components, a hierarchical compression is formulated. The geometry layer captures the rigid structures using physics-aware instancing, while the appearance layer encodes material properties through manifold material quantization.

Stage 3: On-Orbit Assembly. Finally, the compressed data is uploaded to the servicing satellite. A lightweight decoder efficiently assembles the representation of target objects for real-time rendering to provide visual guidance for downstream servicing tasks.

B. Preliminary: 3D Gaussian Splatting

As an explicit radiance field representation, 3DGS models an object as a set of 3D Gaussian primitives $\mathcal{G} = \{g_i\}_{i=1}^N$. Each primitive g_i is parameterized by its center position $\mathbf{p}_i \in \mathbb{R}^3$, covariance matrix $\Sigma_i \in \mathbb{R}^{3 \times 3}$, color $\mathbf{c}_i \in \mathbb{R}^3$, and opacity $\alpha_i \in \mathbb{R}$. The spatial probability density is defined as:

$$G_i(\mathbf{x}) \triangleq \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{p}_i)^\top \Sigma_i^{-1} (\mathbf{x} - \mathbf{p}_i) \right) \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^3$ denotes the position vector of a point in 3D space. The covariance matrix Σ_i controls the shape and orientation of the Gaussian primitive:

$$\Sigma_i = \mathbf{R}_i \mathbf{S}_i \mathbf{S}_i^\top \mathbf{R}_i^\top \quad (3)$$

where $\mathbf{R}_i \in SO(3)$ is a rotation matrix parameterized by a quaternion $\mathbf{q}_i \in \mathbb{R}^4$. $\mathbf{S}_i \in \mathbb{R}^{3 \times 3}$ is a diagonal scaling matrix.

To render the scene from a viewpoint, 3D Gaussians are projected onto the 2D image plane. The 2D covariance Σ_{2D} is approximated via the viewing transformation $\mathbf{W}$ and the Jacobian of the affine projection $\mathbf{J}_i$ as:

$$\Sigma_{2D} = \mathbf{J}_i \mathbf{W} \Sigma_i \mathbf{W}^\top \mathbf{J}_i^\top. \quad (4)$$

The final pixel color $\mathbf{C}$ is computed by blending the N_r ordered primitives:

$$\mathbf{C} = \sum_{i=1}^{N_r} \mathbf{c}_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (5)$$

where $\mathbf{c}_i$ is the view-dependent color modeled by Spherical Harmonics. α_i is the opacity derived from the 2D Gaussian evaluation.

For the optimization of 3DGS, the photometric loss is employed as defined in [3]:

$$L_p = (1 - \gamma_p) L_1 + \gamma_p (1 - L_{SSIM}) \quad (6)$$

where L_1 is the mean absolute error between rendered and ground-truth pixel colors, and L_{SSIM} is the structural similarity index. γ_p balances the two terms.

C. Semantic-Spatial Decomposition

To transition from the unstructured geometric reconstruction to a physics-aware representation, a functional semantic label must be assigned to each Gaussian primitive. This problem can be formally formulated as inverse rendering in the semantic domain: identifying a latent 3D feature field that aligns with the 2D semantic priors extracted from multi-view images when rendered.

To tackle this specific inverse rendering task, a targeted two-stage decoupled optimization strategy is proposed:

- *Geometry-Appearance Training.* The standard 3DGS parameters are optimized using the RGB images captured from the servicing satellite.
- *Semantic Feature Embedding.* With the standard 3DGS parameters fixed, semantic features are learned to facilitate the subsequent decomposition.

The definition of the Gaussian primitive is extended through the attachment of a learnable feature vector $\mathbf{f} \in \mathbb{R}^D$ to the fixed geometric and appearance parameters $\{\mathbf{p}, \mathbf{q}, \mathbf{s}, \alpha, \mathbf{c}\}$. The rendering of this feature field follows the linearity of the physical rasterization process. The rendered feature $\mathbf{F}$ at a pixel is formed by the α -blended summation of the intersecting 3D features:

$$\mathbf{F} = \sum_{i=1}^N \mathbf{f}_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (7)$$

where $\mathbf{f}_i$ is the feature vector of the i -th Gaussian primitive.

To bootstrap structure discovery, unstructured Gaussians are partitioned by lifting 2D semantic priors into the 3D domain. Learnable semantic vectors $\mathbf{f}_i$ are appended to geometrically fixed Gaussian primitives. The Segment Anything Model (SAM) [42] is employed to extract multi-view 2D semantic

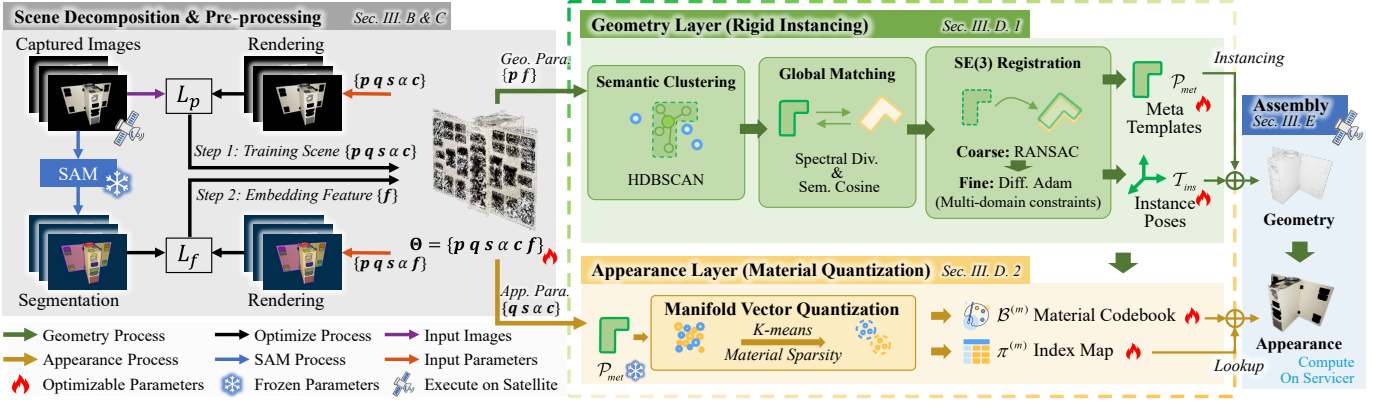


Fig. 3: The overall pipeline of the proposed framework. First, an unstructured Gaussian field is decomposed into semantic components. Then, a hierarchical representation is constructed with a geometry layer and an appearance layer. Finally, the compact representation is packed for efficient storage and real-time rendering.

masks. The ground-truth semantic feature $\mathbf{F}_{gt}$ at each pixel is computed by averaging the encoded semantic labels. The feature rendering loss over M pixels is defined as:

$$L_f = \sum_{i=1}^M \|\mathbf{F}_i - \mathbf{F}_{gt,i}\|_2^2. \quad (8)$$

Equation (7) creates a differentiable bridge between the discrete 3D primitives and continuous 2D feature maps. Equation (8) effectively propagates the 2D semantic consistency into the 3D space through differentiable rasterization, assigning similar feature representations to primitives that spatially project to the same functional component.

D. Hierarchical Representation

Building upon the semantic-spatial decomposition, we reformulate the segmented Gaussian clusters into a hierarchical representation that explicitly separates geometric structure from appearance parameters, reflecting the rigidity and material sparsity inherent to satellite components.

1) *The Geometry Layer*: The geometry layer aims to recover the rigid structures underlying the decomposed component blocks. However, this task is characterized by severe combinatorial ambiguity and non-convex optimization over $SE(3)$. Directly optimizing the poses of all component blocks is prone to suboptimal local minima due to the high dimensionality and non-linearity of the search space. To address this, a coarse-to-fine cascaded strategy is proposed. As illustrated in Table I, the progressive geometry layer reduces the combinatorial search space via invariant-based pair pruning, coarse rigid alignment, and differentiable refinement, with increasing geometric fidelity and computational cost.

Semantic Clustering. To transition from unstructured feature fields to physically meaningful entities, the converged semantic manifold is partitioned into discrete component blocks. Unlike centroid-based algorithms, HDBSCAN is chosen for delineating arbitrary topological structures. This algorithm inherently isolates sparse outliers, serving as a built-in filter to reject view-dependent floaters and optimization noise inherent to raw 3DGS.

TABLE I: Progressive Geometry Layer for Physics-Aware Rigid Instancing. A coarse-to-fine cascade successively prunes implausible matches, initializes rigid alignment, and refines verified instances through differentiable optimization.

Stage	Input Data	Complexity
I. Semantic Clustering	Gaussian Primitives	$\mathcal{O}(D \cdot N^2)$
II. Global Matching	Potential Pairs	$\mathcal{O}(\mathcal{P} ^2)$
III. Coarse Alignment	Plausible Candidates	$\mathcal{O}(\mathcal{K})$
IV. Differentiable Refinement	Verified Instances	$\mathcal{O}(\text{iter} \cdot P_t \log P_s)$

Notation: N is the global primitive count with feature dimension D . $|\mathcal{P}|$ stands for the number of candidate blocks, $|\mathcal{K}|$ for RANSAC correspondences, and $|P_t|, |P_s|$ for the primitive counts of the template and instance.

However, directly applying density-based clustering to millions of primitives entails an intractable $\mathcal{O}(D \cdot N^2)$ computational complexity. To ensure scalability, a core-set assignment strategy is implemented. HDBSCAN distills semantic prototypes from a stochastic subset, followed by a global nearest-prototype assignment. This efficiently aggregates the unstructured primitives into kinematically independent blocks $\mathcal{P} = \{P_1, P_2, \dots, P_k\}$, establishing strict geometric boundaries for the subsequent $SE(3)$ registration.

Global Matching. Given the set of decomposed component blocks $\mathcal{P}$, the first step aims to effectively match blocks in the global search space while retaining plausible kinematic correspondences. Therefore, we seek a rotation-invariant geometric similarity measure that is mathematically well-defined and computable in constant time.

Each block P_k is abstracted as a zero-mean Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{\Lambda}_k)$, where the covariance matrix $\mathbf{\Lambda}_k \in \mathbb{R}^{3 \times 3}$ captures the second-order spatial statistics of its Gaussian centers. Since rigid transformations act on covariances $\mathbf{\Lambda}_k \mapsto \mathbf{R}\mathbf{\Lambda}_k\mathbf{R}^T$, a principled rotation-invariant distance between two blocks P_i and P_j can be defined by minimizing the Kullback-Leibler (KL) divergence over all relative rotations:

$$D_{rot}(P_i, P_j) \triangleq \min_{\mathbf{R} \in SO(3)} \frac{1}{2} \left(D_{kl}(\tilde{\mathcal{N}}_i \| \mathcal{N}_j) + D_{kl}(\mathcal{N}_j \| \tilde{\mathcal{N}}_i) \right) \quad (9)$$

where $\hat{\mathcal{N}}_i = \mathcal{N}(\mathbf{0}, \mathbf{R}\mathbf{\Lambda}_i\mathbf{R}^\top)$ is the rotated distribution of block P_i .

Exploiting the analytic form of KL divergence between Gaussian distributions and applying von Neumann's trace inequality, the minimization in Equation (9) admits a closed-form solution. Specifically, the optimal rotation aligns the principal axes of the two covariances, reducing the invariant divergence to a function solely of their eigenvalue spectra. Denoting the eigenvalues of $\mathbf{\Lambda}_k$ as $\lambda_k = (\lambda_{k,1}, \lambda_{k,2}, \lambda_{k,3})$, the symmetric formulation is adopted to eliminate direction bias, yielding the spectral divergence:

$$D_s(P_i, P_j) = \frac{1}{2} \sum_{m=1}^3 \left(\frac{\lambda_{i,m}}{\lambda_{j,m}} + \frac{\lambda_{j,m}}{\lambda_{i,m}} - 2 \right) \quad (10)$$

where the eigenvalues are sorted in descending order $\lambda_{k,1} \geq \lambda_{k,2} \geq \lambda_{k,3}$.

Importantly, Equation (10) is mathematically equivalent to the rotation-invariant KL divergence in Equation (9), but avoids any explicit optimization over $SO(3)$. Consequently, while the similarity evaluation between any two individual blocks requires only $\mathcal{O}(1)$ constant time using their covariance eigenvalues, the overall computational complexity for exhaustive global pruning is strictly bounded by $\mathcal{O}(|\mathcal{P}|^2)$. This makes it well-suited for screening combinatorial search spaces in large-scale scenarios.

To further suppress false matches between geometrically similar but semantically distinct components, semantic consistency is incorporated into the distance measure. The final distance measure is defined as:

$$D_p(P_i, P_j) \triangleq w_g \mathcal{H}(D_s(P_i, P_j)) + w_s D_f(\bar{\mathbf{f}}_i, \bar{\mathbf{f}}_j) \quad (11)$$

where $\bar{\mathbf{f}}_k$ is the mean semantic feature of block P_k , and $\mathcal{H}(\cdot)$ is a bounded monotonic mapping (implemented as $\tanh$) to stabilize the scale of the geometric term. w_g and w_s are balancing weights. D_f is the cosine distance between the mean semantic features of the two blocks:

$$D_f(\bar{\mathbf{f}}_i, \bar{\mathbf{f}}_j) = 1 - \frac{\bar{\mathbf{f}}_i^\top \bar{\mathbf{f}}_j}{\|\bar{\mathbf{f}}_i\| \|\bar{\mathbf{f}}_j\|}. \quad (12)$$

Only candidate pairs satisfying $D_p(P_i, P_j) < \tau_p$ are regarded as template-instance candidate pairs, which are subsequently passed to the coarse rigid alignment stage.

Coarse Rigid Alignment. The optimization of component-wise rigid poses in 3DGS is highly sensitive to initialization due to the non-convexity of the underlying $SE(3)$ parameterization. Gradient-based methods often fail when facing large rotational discrepancies. To address this, a robust initialization problem is formulated to provide coarse alignment for each candidate template-instance pair.

For a hypothesized template-instance pair derived from the pruning stage, let $P_t \in \mathcal{P}$ denote the template block and $P_s \in \mathcal{P}$ denote the instance block. Each primitive $x \in P_t$ is parameterized by its 3DGS attributes. The Fast Point Feature Histogram (FPFH) [43] descriptor, denoted as $\eta(\cdot)$, is computed over the Gaussian centroids $\mathbf{p}_{x_i}$ to capture local geometric context that is approximately invariant under rigid

motions. Putative correspondences $\mathcal{K}$ are then established by nearest neighbor matching in the FPFH feature space:

$$y(x) = \arg \min_{y' \in P_s} \|\eta(x) - \eta(y')\|_2, \quad \forall x \in P_t, \quad (13)$$

$$\mathcal{K} = \{(x, y(x)) \mid x \in P_t\}. \quad (14)$$

A coarse rigid transformation $\mathbf{T}_{init} = [\mathbf{R}|\mathbf{t}] \in SE(3)$ is subsequently estimated by maximizing the geometric consensus over $\mathcal{K}$. The consensus objective is expressed as:

$$\mathbf{T}_{init} = \arg \max_{\mathbf{T} \in SE(3)} \sum_{(x,y) \in \mathcal{K}} \mathbb{I}(\|\mathbf{R}\mathbf{p}_x + \mathbf{t} - \mathbf{p}_y\|_2 < \tau_c) \quad (15)$$

where $\mathbb{I}(\cdot)$ is the indicator function and τ_c is a distance threshold. This maximization is efficiently approximated using RANSAC, yielding an initialization that places the estimated pose within the region of convergence of the subsequent differentiable refinement stage. Its computational cost scales linearly with the size of the putative feature correspondence set $|\mathcal{K}|$ (i.e., $\mathcal{O}(|\mathcal{K}|)$), making it far sparser in complexity than dense point cloud methods.

Differentiable Refinement. Starting from the coarse initialization $\mathbf{T}_{init}$, the subsequent refinement aims to achieve high-fidelity alignment. The most straightforward approach is to directly optimize the rigid transformation $\mathbf{T}$. However, because of the intrinsic distributional divergence of discrete 3DGS primitive blocks, a strictly rigid $SE(3)$ mapping induces severe photometric degradation, as demonstrated by empirical results in Section IV-C. A 6-DoF isometric transformation exhibits a severe representational capacity deficit, lacking the ability to compensate for local illumination variations, view-dependent artifacts, and optimization-induced noise. Therefore, to achieve high-fidelity rendering without sacrificing the compactness of template instantiation, it is necessary in practice to reshape the internal Gaussian parameters Θ while preserving the underlying rigid transformation $\mathbf{T}$.

Furthermore, a naive approach to resolving this representational deficit is jointly optimizing Θ and $\mathbf{T}$, which leads to an unconstrained optimization problem with a significantly enlarged search space. The 2D photometric rendering loss tends to scatter primitives uncontrollably to compensate for local projection errors, while 3D topological constraints demand strict geometric convergence. To break this optimization deadlock, the differentiable refinement is decoupled into two orthogonal alternating processes: updating the intrinsic Gaussian parameters Θ and refining the extrinsic rigid transformation $\mathbf{T}$.

First, at the k -th cycle ($k \in \{1, \dots, K_r\}$), the intrinsic parameters Θ are frozen, and the rigid transformation $\mathbf{T}$ is optimized to minimize the photometric loss and depth consistency:

$$\mathbf{T}_k = \arg \min_{\mathbf{T} \in SE(3)} (L_p(\mathbf{T}, \Theta_{k-1}) + \gamma_d L_d(\mathbf{T}, \Theta_{k-1})) \quad (16)$$

where L_p is the photometric loss defined in Equation (6), and L_d is a depth consistency term. Both losses are evaluated by projecting the transformed template block into the image

plane, computing the pixel-wise color and geometric discrepancies against the target instance's ground-truth renderings. γ_d is a balancing weight.

Crucially, this pose optimization also serves as a verification gate. To filter out spurious matches generated during the coarse alignment, a candidate pair is retained for subsequent alternating refinement only if it satisfies the following criteria after the initial optimization iteration:

$$L_p(\mathbf{T}_1, \Theta_0) + \gamma_d L_d(\mathbf{T}_1, \Theta_0) < \tau_r \quad (17)$$

where $\mathbf{T}_1$ is the optimized transformation after the first iteration, and τ_r is a predefined threshold. This strict gating ensures that the subsequent manifold reshaping is executed exclusively on topologically compatible instances, thereby preventing geometric collapse. All successfully verified matches are collected into the set:

$$\mathcal{M} = \{(P_{t,k}, P_{s,k}) \mid k = 1, 2, \dots, K\} \quad (18)$$

where $P_{t,k}$ and $P_{s,k}$ are the template and instance blocks of the k -th verified match.

Subsequently, the transformations $\mathbf{T}$ are frozen. All parameters Θ are updated by jointly minimizing a global photometric constraint and a set of local block-wise distribution constraints:

$$\Theta_k = \arg \min_{\Theta} \left(\gamma_{\theta} L_p(\mathbf{T}_{k-1}, \Theta) + \sum_{m \in \mathcal{M}} L_{l,m}(\mathbf{T}_{k-1}, \Theta) \right) \quad (19)$$

where $L_p(\mathbf{T}_{k-1}, \Theta)$ is the photometric error of the fully re-assembled scene, rendered by applying transformations $\mathbf{T}_{k-1}$ to the instantiated template components against the ground truth. For a given matched pair $m = (P_t, P_s)$, the local consistency objective $L_{l,m}$ is decomposed across three parameter domains:

$$L_l = L_c + \gamma_{cd} L_{cd} + \gamma_s L_s \quad (20)$$

where γ_{cd} and γ_s are balancing weights.

L_{cd} is a Chamfer distance that regularizes the spatial topology of the matched blocks:

$$L_{cd} = \frac{1}{|P_t|} \sum_{x \in P_t} \min_{y \in P_s} \|\hat{\mathbf{p}}_x - \mathbf{p}_y\|_2^2 + \frac{1}{|P_s|} \sum_{y \in P_s} \min_{x \in P_t} \|\hat{\mathbf{p}}_x - \mathbf{p}_y\|_2^2 \quad (21)$$

where $\hat{\mathbf{p}}_x = \mathbf{R}\mathbf{p}_x + \mathbf{t}$ is the transformed position of a template primitive $x \in P_t$.

L_c and L_s are Mean Squared Error (MSE) constraints that enforce consistency in color and scale manifolds, respectively:

$$L_c = \frac{1}{|P_t|} \sum_{x \in P_t} \|\mathbf{c}_x - \mathbf{c}_{\hat{y}(x)}\|_2^2 + \frac{1}{|P_s|} \sum_{y \in P_s} \|\mathbf{c}_{\hat{x}(y)} - \mathbf{c}_y\|_2^2 \quad (22)$$

$$L_s = \frac{1}{|P_t|} \sum_{x \in P_t} \|\mathbf{s}_x - \mathbf{s}_{\hat{y}(x)}\|_2^2 + \frac{1}{|P_s|} \sum_{y \in P_s} \|\mathbf{s}_{\hat{x}(y)} - \mathbf{s}_y\|_2^2 \quad (23)$$

where $\hat{x}(y)$ and $\hat{y}(x)$ are the nearest neighbor correspondences derived from the minimization operations in Equation (21).

These constraints induce stable and well-conditioned gradients that guide the rigid transformation precisely. While computationally intensive with a complexity of $\mathcal{O}(\text{iter} \cdot |P_t| \log |P_s|)$, this step is strictly confined to the few verified instances, preserving overall system efficiency.

2) *The Appearance Layer*: While the geometry layer re-constructs the rigid structure, the appearance layer focuses on encoding the material properties of the target object. For aerospace objects, appearance variations are largely driven by a limited set of surface materials and structural components (*e.g.*, metallic frames, solar panels, thermal coatings), leading to highly repetitive and low-diversity appearance patterns across Gaussians.

Specifically, the rotation of individual Gaussian primitives $\mathbf{q}$ effectively acts as a surface texture property [44], similar to micro-facet normals in BRDF models, rendering it highly amenable to the same manifold quantization strategy used for color and opacity. Therefore, the appearance layer employs a unified manifold quantization scheme to compress properties $\Theta_{app} = \{\mathbf{q}, \mathbf{s}, \alpha, \mathbf{c}\}$.

Motivated by the material-inspired regularities observed in aerospace targets, an independent component-wise quantization is adopted. For a given appearance attribute $\theta \in \Theta_{app}$, a learnable codebook is constructed:

$$\mathcal{B}^{(m)} = \left\{ \mathbf{b}_k^{(m)} \in \mathbb{R}^{d_m} \mid k = 1, \dots, K_m \right\} \quad (24)$$

where K_m is the number of codewords for attribute m , and d_m is the dimensionality of each codeword. $\mathbf{b}_k^{(m)}$ represents the k -th codeword for attribute m .

Given the continuous parameter vector $\theta_i^{(m)}$ of the i -th Gaussian primitive, a discrete index map $\pi^{(m)} : \{1, 2, \dots, N_p\} \rightarrow \{1, 2, \dots, K_m\}$ is derived via nearest-neighbor assignment:

$$\pi^{(m)}(i) = \arg \min_k \left\| \theta_i^{(m)} - \mathbf{b}_k^{(m)} \right\|_2 \quad (25)$$

where $N_p = |P_t|$ is the total number of Gaussian primitives in the meta templates.

During rendering, the original appearance parameter is replaced by its corresponding prototype $\mathbf{b}_{\pi^{(m)}(i)}^{(m)}$. This independent quantization strategy avoids mixing heterogeneous attributes with different physical meanings and sensitivity profiles, while still effectively capturing the low-dimensional structure of the appearance space.

Overall, the proposed appearance layer exploits material-inspired regularities through independent prototype-based quantization, enabling efficient compression while preserving visual fidelity and full compatibility with the original 3D Gaussian Splatting framework.

The derived geometry and appearance compact manifolds constitute a unified optimization framework tailored for aerospace objects. For clarity and reproducibility, the explicit mapping of the mathematical derivations discussed above to the computational steps is formally specified in Algorithm 1.

E. On-Orbit Assembly

The on-orbit assembly is the inverse process of the hierarchical compression described above. The compressed representation $\{\mathcal{P}_{met}, \mathcal{T}_{ins}, \mathcal{B}^{(m)}, \pi^{(m)}\}$ is transmitted to the servicing satellite. The target 3DGS is reconstructed by instantiating templates at the estimated poses and retrieving appearance parameters from the corresponding codebooks.

Algorithm 1 Physics-Aware Hierarchical 3DGS Compression

Require: Images $\mathcal{I} = \{I_k\}_{k=1}^K$, Camera Poses $\mathcal{T} = \{\mathbf{T}_k\}_{k=1}^K$
Ensure: Hierarchical Compact 3DGS $\hat{\mathcal{G}} = \{\mathcal{G}_{geo}, \mathcal{G}_{app}\}$

```

/* Stage 1: Pre-processing */
1:  $\mathcal{G} \leftarrow \arg \min_{\mathcal{G}} L_p$  // Eq. (5), (6)
2:  $\mathbf{F} \leftarrow \arg \min_{\mathbf{F}} L_f$  // Eq. (7), (8)
/* Stage 2: Hierarchical Compression */
3: // Geometry Layer
4:  $\mathcal{P} \leftarrow \{P_1, \dots, P_k\} \leftarrow \text{Cluster}(\mathcal{G}, \mathbf{F})$  // HDBSCAN
5:  $\mathcal{M} \leftarrow \emptyset$  // Candidate template-instance pairs set
6: for  $(P_i, P_j) \in \mathcal{P} \times \mathcal{P}$  do // Global matching
7:    $D_s(P_i, P_j) \leftarrow \text{Equation (10)}(\lambda_i, \lambda_j)$ 
8:    $D_p(P_i, P_j) \leftarrow \text{Equation (11)}(D_s(P_i, P_j), \bar{\mathbf{f}}_i, \bar{\mathbf{f}}_j)$ 
9:   if  $D_p(P_i, P_j) < \tau_p$  then  $\mathcal{M} \leftarrow \mathcal{M} \cup \{(P_i, P_j)\}$ 
10:  end if
11: end for
12: for  $(P_t, P_s) \in \mathcal{M}$  do // Coarse alignment
13:    $\mathcal{K} \leftarrow (x, y(x))$  // FPFH matching (Eq. (13), (14))
14:    $\mathbf{T}_{init} \leftarrow \text{Equation (15)}(\mathcal{K})$ 
15: end for
16: for  $k = 1$  to  $K_r$  do // Differentiable refinement
17:   for  $(P_t, P_s) \in \mathcal{M}$  do
18:      $\mathbf{T}_k \leftarrow \arg \min_{\mathbf{T}} (L_p + \gamma_d L_d)$  // Eq. (16)
19:     if  $k == 1$  and  $L_p + \gamma_d L_d > \tau_r$  then
20:        $\mathcal{M} \leftarrow \mathcal{M} \setminus \{(P_t, P_s)\}$  // Eq. (18)
21:     end if
22:   end for
23:    $\Theta_k \leftarrow \arg \min_{\Theta} (\gamma_{\theta} L_p + \sum_{m \in \mathcal{M}} L_{l,m})$  // Eq. (19)
24: end for
25:  $\mathcal{G}_{geo} \leftarrow \text{Instance}(\mathcal{P}_{met}, \mathcal{T}_{ins})$ 
26: // Appearance Layer
27:  $\mathcal{B}^{(m)} \leftarrow \text{K-means}(\{\theta_i^{(m)}\}_{i=1}^N)$  // Eq. (24)
28:  $\pi^{(m)} \leftarrow \text{Equation (25)}$ 
29: return  $\hat{\mathcal{G}} = \{\mathcal{G}_{geo}, \mathcal{G}_{app}\}$ 

```

To reconstruct an instance from its associated template, the global state $(\hat{\mathbf{p}}_s, \hat{\Sigma}_s)$ of each primitive is instantiated from its local template parameters $(\mathbf{p}_t, \Sigma_t)$ via the forward kinematic transformation:

$$\hat{\mathbf{p}}_s = \mathbf{R}\mathbf{p}_t + \mathbf{t}, \quad \hat{\Sigma}_s = \mathbf{R}\Sigma_t\mathbf{R}^T \quad (26)$$

where $\mathbf{T} = [\mathbf{R}|\mathbf{t}] \in SE(3)$ is the transformation matrix retrieved from the stored pose set $\mathcal{T}_{ins}$.

Simultaneously, the appearance attributes $\hat{\theta}_i^{(m)}$ are decoded with the quantization look-up table:

$$\hat{\theta}_i^{(m)} = \mathbf{b}_{\pi(i)}^{(m)}, \quad \mathbf{b}^{(m)} \in \mathcal{B}^{(m)}. \quad (27)$$

The final scene is the union of the static templates and the reconstructed instances using Equations (26) and (27).

The proposed representation involves a series of highly intractable operations, including combinatorial clustering with $\mathcal{O}(D \cdot N^2)$, global matching with $\mathcal{O}(|\mathcal{P}|^2)$, and differentiable refinement with $\mathcal{O}(\text{iter} \cdot |P_t| \log |P_s|)$. This cumulative computational bottleneck mathematically mandates our asymmetric space-ground paradigm. By completely offloading the entire

scene training and hierarchical compression to the ground station, the on-orbit computational burden is strictly bounded to linear time $\mathcal{O}(N)$ for deterministic manifold retrieval and affine instantiation, ensuring real-time perception under severe hardware constraints.

IV. EXPERIMENTS AND ANALYSIS

Embedding physical priors into 3DGS can fundamentally overcome the rate-distortion bottleneck of signal-level compression. To validate this, we conducted comprehensive evaluations on high-fidelity synthetic and real-world aerospace datasets. First, the dataset collection and experimental setup are described. Then, our method is systematically benchmarked against state-of-the-art compression baselines. The experiments were specifically designed to assess overall rate-distortion superiority, the preservation of structural integrity at extremely low bitrates, and the effectiveness of each component through ablation studies.

A. Experimental Settings

1) *Dataset:* Evaluating physics-aware compression for aerospace targets requires benchmarks that faithfully capture the structural repetition and material scarcity of industrial spacecraft, which are absent in general datasets such as NeRF Synthetic [2] or Deep Blending [45]. As part of our ongoing efforts to bridge the gap in space visual perception, our previously established Spacecraft Pose Estimation and 3D Reconstruction Dataset (SPARD) [11] is leveraged for this evaluation. Unlike existing aerospace datasets that are constrained by limited scale, SPARD encompasses 37 highly detailed spacecraft models sourced from National Aeronautics and Space Administration (NASA) [46], European Space Agency (ESA) [47] and a flight-proven CubeSat provided by Smart Satellite Technology Co., Ltd. These targets are characterized by intricate topological structures with highly rigid repetitive patterns and low material diversity, making them an ideal benchmark for validating our core hypothesis regarding component instancing and material quantization.

Crucially, to evaluate the sim-to-real transferability of the proposed method, SPARD consists of two subsets: high-fidelity synthetic renderings generated via Unreal Engine 5 (UE5), and real-world sequences captured through two Universal Robots UR5 arms using ZED 2i sensors. This sim-to-real diversity ensures that the method's performance is not confined to idealized conditions, but extends to the complexities of real-world space imagery. Figure 4 illustrates the set of synthetic and real scenes used in the experiments.

2) *Metrics:* Three standard metrics are adopted to evaluate the rendering quality of the proposed method and baselines: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) [48]. For structural compactness, the absolute cost is quantified by the physical storage memory size in megabytes (MB) and the total retained primitive count.

Furthermore, to systematically benchmark the relative compression efficiency against the uncompressed baseline, two

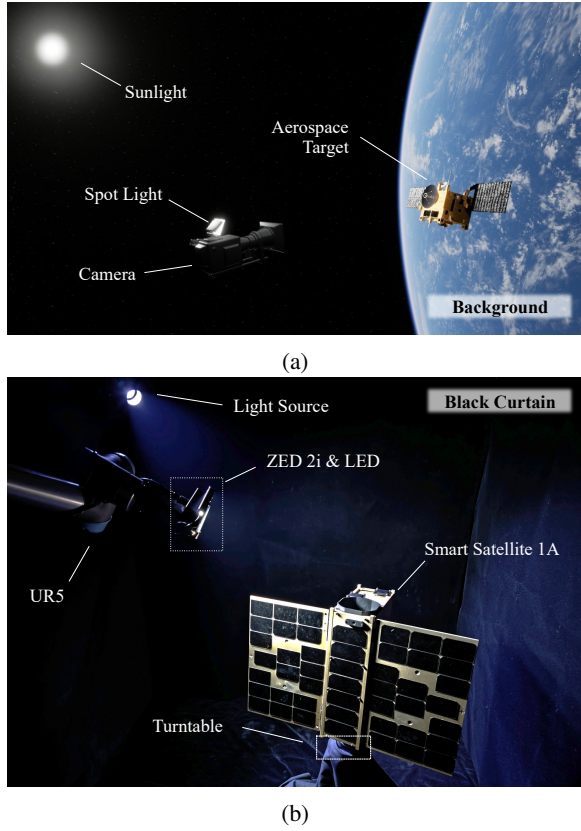


Fig. 4: The key components of the SPARD dataset [11]. (a) Synthetic renderings of spacecraft models with UE5. (b) Real-world captures of physical spacecraft mock-ups, exhibiting realistic sensor noise and lighting variations.

relative Compression Ratio (CR) metrics are defined. The Primitive Compression Ratio (CR_p) is defined as:

$$CR_p = N_{orig}/N_{inst} \quad (28)$$

where N denotes the primitive count. N_{inst} and N_{orig} denote the primitive count of the compressed instance-based representation and the original uncompressed 3DGS, respectively. This metric directly validates the efficacy of the pruning effect of the proposed geometric layer.

The Storage Compression Ratio (CR_s) is defined as:

$$CR_s = S_{orig}/S_{final} \quad (29)$$

where S represents the total memory size. S_{final} and S_{orig} denote the storage memory size of the final compressed representation and the original uncompressed 3DGS, respectively. This metric reflects the overall compression performance, encompassing the gains from both structural sharing and physical material quantization.

3) *Parameters*: Following the previous 3DGS work [38], the standard 3DGS was pre-trained for 10,000 iterations. The dimensionality of the semantic feature vector $\mathbf{f}$ in Equation (7) was set to $D = 32$. During the preliminary photometric optimization in Equation (6), the weight γ_p was set to 0.2.

In the geometry layer, the balancing weights for geometric and semantic distances in Equation (11) were set to $w_g = 0.4$

and $w_s = 0.6$, respectively. The pruning threshold was defined as $\tau_p = 0.8$. For the coarse rigid alignment, the distance threshold τ_c in Equation (15) was adaptively set to 2% of the maximum extent of the target bounding box. In the differentiable refinement stage, the rejection threshold τ_r in Equation (17) was set to 0.005. The balancing weights for the refinement losses in Equation (19) were set to $\gamma_\theta = 10$, $\gamma_{cd} = 10$, $\gamma_c = 1.0$, and $\gamma_s = 2$. The alternating optimization was executed for $K_r = 7$ cycles. Within each cycle, the transformation $\mathbf{T}$ was optimized for 1,000 iterations in the initial cycle and 150 iterations in subsequent cycles, while the parameters Θ were updated for 5,000 iterations per cycle, which was extended to 10,000 iterations in the final cycle to strictly enforce photometric fidelity.

In the appearance layer, the codebook size in Equation (24) was uniformly configured to $K_m = 4096$ across all independent appearance attributes.

All experiments were conducted on a desktop equipped with an AMD EPYC 7T83 CPU and an NVIDIA RTX 4090 GPU.

B. Performance Evaluation

To validate the generalizability and robustness of the proposed method, comprehensive evaluations were conducted on both synthetic and real-world subsets of the SPARD dataset. While SPARD encompasses 37 models, following the standard evaluation protocol of novel view synthesis paradigms (e.g., NeRF [2], 3DGS [3]), a representative subset was selected for detailed analysis. The evaluated satellite models, as illustrated in Fig. 6 (a), exhibit a typical aerospace design characterized by a central bus, solar panels, and other equipment. These models are ideal for validating the efficacy of component instancing and material quantization.

Table II details the quantitative compression performance across these targets. Despite the differences in structural and material composition, our physics-aware compression framework demonstrates remarkable robustness. Across all evaluated models, the proposed method achieves significant reductions in primitive count, with CR_p of at least $1.15\times$, while maintaining high rendering fidelity with PSNR values above 30 dB, SSIM values above 0.97 and LPIPS values below 0.04. The overall storage compression ratio CR_s is over $5.5\times$, which highlights the joint advantages of structural sharing and material quantization.

Having validated the robustness of our method across diverse aerospace targets, we further benchmarked the rate-distortion performance against the original 3DGS [3] and recent state-of-the-art compression methods, including pruning-based methods like LightGaussian [20], vector quantization methods like CompGS [23] and anchor-based methods like Scaffold-GS [26]. EAGLES [25] and CompGS [23] are typical comprehensive compression methods. The results are summarized in Table III, which systematically compares the average rate-distortion performance of our method against state-of-the-art compression schemes across evaluated subsets.

As shown in Table III and Fig. 5, existing methods exhibit different trade-offs between reconstruction quality and storage efficiency. As the uncompressed baseline, the vanilla

TABLE II: Quantitative performance of the proposed physics-aware hierarchical compression method on the SPARD dataset. The configuration of the full compression includes both the geometry and appearance layers, which is denoted as “w/ App.” in the table.

Object	Stage	PSNR $\uparrow$ (dB)	SSIM $\uparrow$	LPIPS $\downarrow$	Points $\downarrow$	CR $_p$	Size $\downarrow$ (MB)	CR $_s$
SS-1A (Real)	w/o Comp.	41.04	0.995	0.011	77346	1.00 $\times$	17.41	1.00 $\times$
	w/o Diff.	22.48	0.927	0.070	50931	1.52 $\times$	11.46	1.52 $\times$
	w/o Alter.	33.74	0.989	0.018	62863	1.23 $\times$	14.14	1.23 $\times$
	w/ Geo.	38.36	0.992	0.016	62863	1.23 $\times$	14.14	1.23 $\times$
	w/ App. (Full)	35.28	0.986	0.022	62863	1.23 $\times$	2.20	7.91 $\times$
SS-1A (Virtual)	w/o Comp.	43.32	0.997	0.010	106137	1.00 $\times$	23.89	1.00 $\times$
	w/o Diff.	14.81	0.882	0.114	65359	1.62 $\times$	14.71	1.18 $\times$
	w/o Alter.	24.87	0.974	0.037	90432	1.17 $\times$	20.35	1.17 $\times$
	w/ Geo.	38.39	0.994	0.022	90432	1.17 $\times$	20.35	1.17 $\times$
	w/ App. (Full)	33.91	0.979	0.033	90432	1.17 $\times$	2.83	8.44 $\times$
Proba-2	w/o Comp.	40.43	0.992	0.012	32506	1.00 $\times$	7.32	1.00 $\times$
	w/o Diff.	23.72	0.950	0.056	16161	2.01 $\times$	3.64	4.79 $\times$
	w/o Alter.	29.79	0.962	0.050	18893	1.72 $\times$	4.25	1.72 $\times$
	w/ Geo.	38.53	0.991	0.015	18893	1.72 $\times$	4.25	1.72 $\times$
	w/ App. (Full)	34.95	0.982	0.022	18893	1.72 $\times$	1.26	5.81 $\times$
Venus Express	w/o Comp.	34.92	0.991	0.015	86015	1.00 $\times$	19.36	1.00 $\times$
	w/o Diff.	17.39	0.857	0.155	40818	2.11 $\times$	9.19	1.90 $\times$
	w/o Alter.	25.42	0.966	0.038	74242	1.16 $\times$	16.71	1.16 $\times$
	w/ Geo.	32.91	0.986	0.020	74242	1.16 $\times$	16.71	1.16 $\times$
	w/ App. (Full)	30.80	0.974	0.030	74242	1.16 $\times$	2.48	7.81 $\times$
Acrimsat	w/o Comp.	44.68	0.998	0.002	71646	1.00 $\times$	16.13	1.00 $\times$
	w/o Diff.	19.12	0.923	0.077	34296	2.09 $\times$	7.72	2.26 $\times$
	w/o Alter.	33.02	0.994	0.005	52755	1.36 $\times$	11.87	1.36 $\times$
	w/ Geo.	40.26	0.997	0.002	52755	1.36 $\times$	11.87	1.36 $\times$
	w/ App. (Full)	36.43	0.993	0.005	52755	1.36 $\times$	1.98	8.13 $\times$

TABLE III: Quantitative Comparison of Rate-Distortion Performance against baselines. The **best** score, **second-best** score, and **third-best** score of compression metrics are highlighted respectively.

Method	PSNR $\uparrow$ (dB)	SSIM $\uparrow$	LPIPS $\downarrow$	Points $\downarrow$	CR $_p$	Size $\downarrow$ (MB)	CR $_s$
3DGS [3]	40.88	0.995	0.010	74730	1.00 $\times$	16.82	1.00 $\times$
2DGS [32]	35.43	0.993	0.013	37459	1.99 $\times$	8.29	2.03 $\times$
EAGLES [25]	33.67	0.983	0.024	54949	1.36 $\times$	2.26	7.44 $\times$
GSCoDec [49]	34.26	0.985	0.016	58755	1.27 $\times$	2.34	7.19 $\times$
LightGaussian [20]	35.00	0.988	0.015	24979	2.99 $\times$	6.41	2.62 $\times$
Scaffold-GS [26]	41.88	0.996	0.007	32300	2.31 $\times$	8.46	1.99 $\times$
CompGS [23]	35.53	0.991	0.017	74730	1.00 $\times$	2.46	6.84 $\times$
Ours	34.27	0.982	0.022	59837	1.25 $\times$	2.15	7.82 $\times$

3DGS [3] inherently defines the upper bound for visual fidelity with PSNR = 40.88 dB and the raw storage baseline of 16.82 MB, against which all compression paradigms are measured. 2DGS [32] improves perceptual quality with LPIPS = 0.013 while reducing the primitive count by about half (CR $_p$ = 1.99 $\times$), but its overall storage remains relatively high at 8.29 MB. LightGaussian [20] further reduces the number of Gaussians (CR $_p$ = 2.99 $\times$), but the resulting model size is still 6.41 MB. As a representative of quantization-centric methods, CompGS [23] achieves more compact representations (2.46 MB), but redundancy in the primitive structure is not explicitly addressed (CR $_p$ = 1.00 $\times$).

In contrast, the proposed method jointly exploits geometric

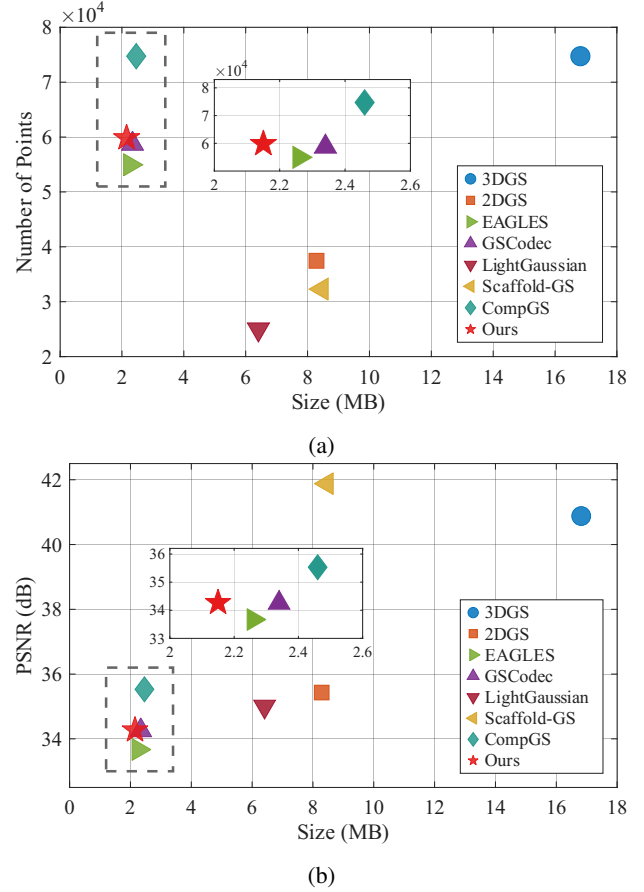


Fig. 5: Quantitative comparison of structural efficiency and rate-distortion against baselines. (a) compares the primitive count and storage size across methods. (b) shows the PSNR values and demonstrates the rate-distortion performance.

instancing and parameter quantization, leading to a compact representation of 2.15 MB (CR $_s$ = 7.82 $\times$). Notably, under comparable reconstruction quality, our method requires less storage while maintaining a similar number of effective primitives compared to GSCoDec [49], resulting in an 8.8% reduction in memory usage.

Figure 6 visually corroborates the rate-distortion analysis, presenting rendering results across diverse aerospace targets. Despite the minor rendering quality degradation discussed previously, the overall visual fidelity of our compressed representation is comparable with the ground truth and larger storage baselines. This confirms that the extreme payload reduction does not incur catastrophic visual degradation, satisfying the requirements for teleoperation and on-orbit visual perception.

C. Ablation Analysis

1) *Synergy of the Hierarchical Architecture*: The incremental rate-distortion impacts of the Geometry and Appearance Layers are reported in Table II to validate our architectural design.

Introducing the geometry layer (w/ Geo.) significantly reduces the number of primitives by grouping repetitive structures into shared templates. For example, in the Proba-2 model,

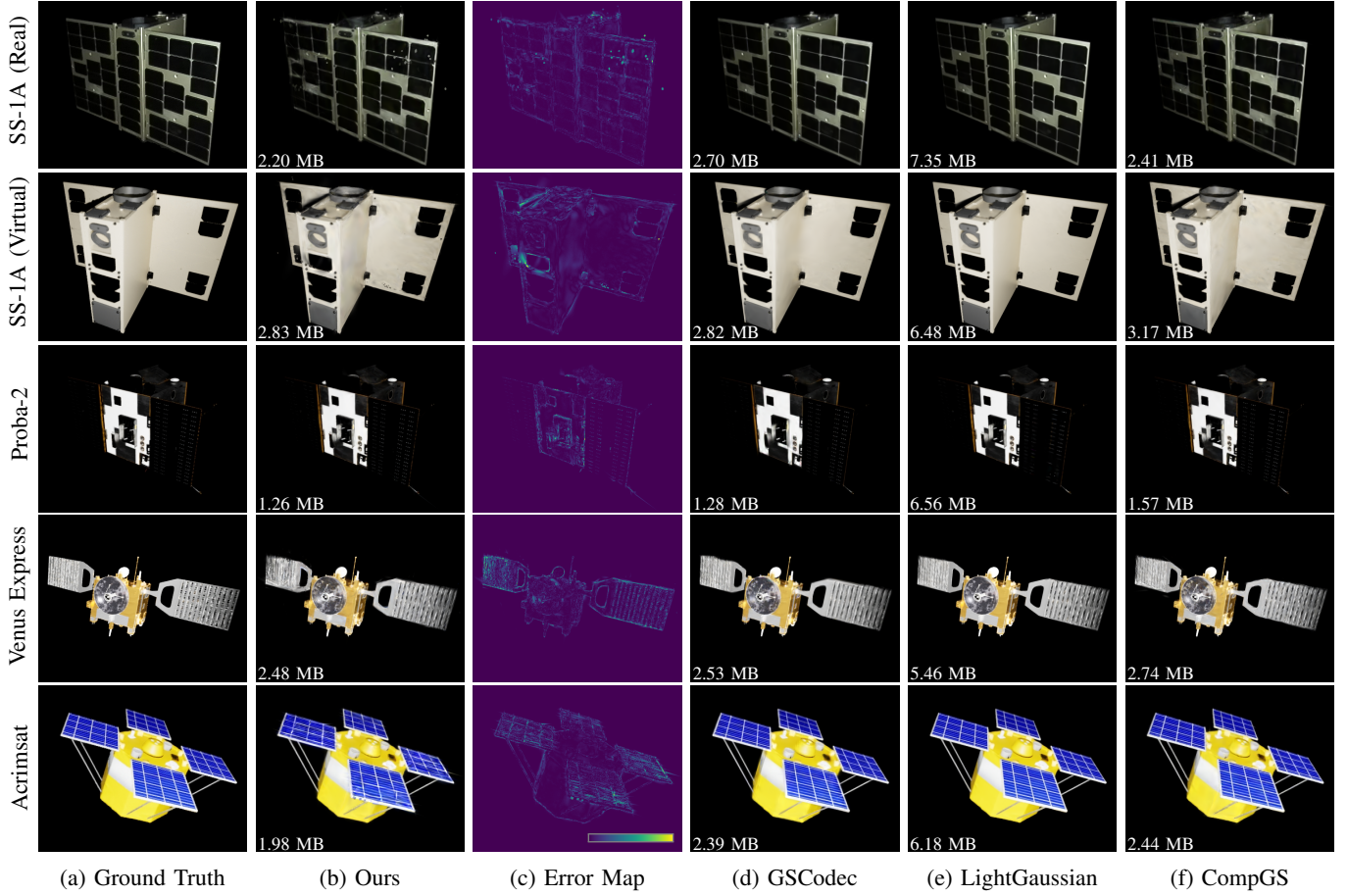


Fig. 6: Qualitative comparison of rendering quality against baselines. The first column shows the ground truth renderings, while the second and third columns show the results of our method and the corresponding error maps. The fourth column shows the result of GSCoDec [49], a representative of methods that jointly optimize pruning and quantization. The last two columns compare the visual quality of our method against LightGaussian [20] and CompGS [23], which represent pruning-based and quantization-based baselines, respectively.

the primitive count decreases from 32,506 to 18,893, yielding a storage compression ratio of $1.72\times$. Since per-primitive precision remains unchanged, the storage reduction directly follows the decrease in primitive count. As expected, enforcing rigid kinematic coherence induces a minor photometric drop. The PSNR decreases from 40.43 dB to 38.53 dB, but the overall visual quality remains high, with SSIM above 0.98 and LPIPS below 0.03.

The appearance layer (w/ App.) subsequently performs material-aware vector quantization. This operation functions entirely within the numerical domain, preserving the exact primitive count extracted by the geometry layer. For all evaluated models, the storage size is further reduced by approximately $5\times$ compared to the geometry layer alone, resulting in an overall compression ratio of over $5.5\times$ relative to the uncompressed baseline.

These results indicate that structural redundancy and parameter redundancy are complementary. By jointly reducing the primitive count and parameter precision, the proposed framework achieves up to $5.5\times$ compression while maintaining high rendering quality with a PSNR exceeding 30 dB.

2) *Efficacy of Differentiable Refinement*: Table II also isolates the photometric contribution of the progressive differentiable refinement mechanism. To accurately evaluate geometric and rendering fidelity without the compounding effects of material quantization, these variants were benchmarked against the full Geometry Layer (w/ Geo.).

Bypassing the refinement entirely (w/o Diff.) relies exclusively on the PPFH and RANSAC-derived initialization. As observed across all scenes, this pure spatial alignment induces severe photometric degradation. For example, in the virtual scene of SS-1A, the PSNR plummets from 38.39 dB to 14.81 dB. Since discrete Gaussian primitives inherently possess stochastic distributional variances generated during the initial unconstrained optimization, macro-level point cloud registration is fundamentally inadequate for pixel-level rasterization fidelity.

Gradient-driven alignment without alternating optimization (w/o Alter.) provides limited visual improvements. While the PSNR of Venus Express increases from 17.39 dB to 25.42 dB, it remains inferior to the full geometry layer at 32.91 dB. This large performance gap supports our claim: a 6-DoF isometric transformation lacks the capacity to compensate for local

illumination differences and non-isotropic shape discrepancies between the template and the target instance.

3) *Trade-offs in Geometric Strictness and Material Capacity*: The proposed hierarchical framework relies on two principal hyperparameters to explicitly navigate the Rate-Distortion trade-off: the pruning rejection threshold τ_r in the geometry layer and the codebook size K_m in the appearance layer.

Geometric Strictness τ_r . Physically, τ_r dictates the strictness of the component matching. We ablated τ_r from 0.003 to 0.011 across diverse targets to analyze its impact. The quantitative relationships among the threshold, storage compression ratio, and rendering quality are illustrated in Fig. 7.

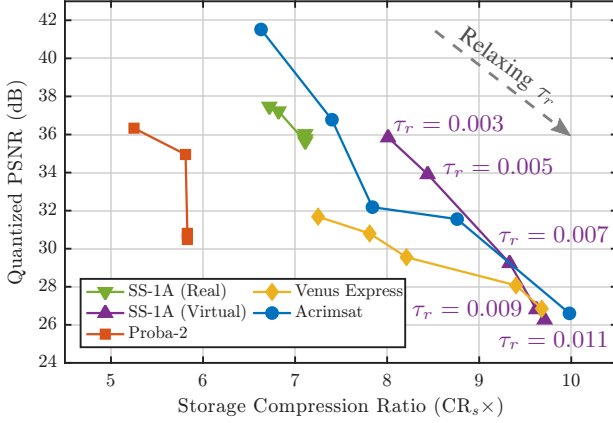


Fig. 7: Ablation on the pruning rejection threshold τ_r in Equation (17). The plot shows the trade-off between compression ratio and reconstruction quality as τ_r is varied.

As the threshold is relaxed from $\tau_r = 0.003$ to $\tau_r = 0.011$, the operating points consistently move toward higher storage compression ratios but lower quantized PSNR. This confirms that a smaller τ_r enforces conservative structural matching, preserving reconstruction quality at the cost of limited redundancy removal. For instance, our method achieves a PSNR of 41.51 dB at $\tau_r = 0.003$ on the Acrimsat model, but its PSNR decreases to 26.60 dB when $\tau_r = 0.011$, as more geometrically inconsistent components are forced to share templates. Similar trends are observed on other targets, confirming the generality of this trade-off across different structural complexities and material compositions.

In particular, $\tau_r = 0.005$ lies near the knee of the curves for most tested assets: it removes substantial structural redundancy while avoiding the severe quality drop observed at larger thresholds. For the real scene of SS-1A, this setting achieves 35.28 dB quantized PSNR with a $7.91\times$ payload reduction. τ_r was therefore set to 0.005 as the default value in all experiments.

Appearance Capacity K_m . While τ_r controls the structural sparsity, the codebook size K_m bounds the representational capacity of the numerical material manifold. We ablated K_m from 512 to 8192 to determine the optimal quantization granularity. The resulting rate-distortion curves are shown in Fig. 8.

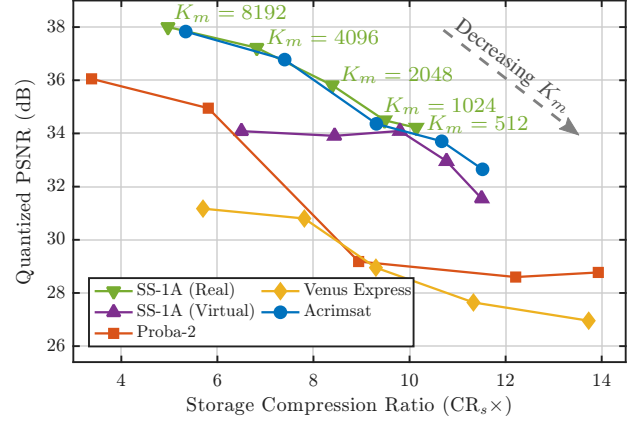


Fig. 8: Ablation on the codebook size K_m in Equation (24). The plot shows the trade-off between compression ratio and reconstruction quality as K_m is varied.

Across all targets, increasing K_m from 512 to 8192 raises the average quantized PSNR from 30.83 dB to 35.43 dB, while reducing the average storage compression ratio from $12.16\times$ to $5.18\times$. This confirms that a larger codebook allows for finer-grained material representation, improving visual fidelity at the cost of increased storage. For instance, on the Proba-2 model, $K_m = 512$ achieves a PSNR of 28.77 dB with a $13.93\times$ compression ratio, while $K_m = 8192$ achieves 36.05 dB with a $3.38\times$ compression ratio. This trade-off holds generally for all targets.

As shown in Fig. 8, from $K_m = 512$ to $K_m = 4096$, the average PSNR increases by only 0.70 dB, whereas the average compression ratio drops by about $2.08\times$. This indicates that overly large codebooks preserve increasingly fine distinctions that bring limited reconstruction benefit. $K_m = 4096$ was therefore set as the default setting to balance reconstruction fidelity and storage reduction.

V. CONCLUSION

In this paper, we presented a physics-aware hierarchical 3DGS compression framework to address the stringent bandwidth and storage constraints of space-ground communications. By recognizing aerospace targets as highly structured industrial assemblies rather than unstructured point clouds, we decoupled the unstructured Gaussian field into a kinematic geometry layer and a quantized appearance layer. Our method leverages a coarse-to-fine cascade strategy to securely map repetitive components to shared rigid templates, strictly preserving topological integrity. The appearance layer then adopts vector quantization, which effectively constrains the capacity of the material manifold.

Our comprehensive evaluations validate that shifting the compression paradigm toward semantic-level component matching substantially alleviates the rate-distortion bottleneck inherent in traditional signal coding. By jointly utilizing structural and parameter redundancy, our approach achieves substantial payload compression down to roughly 2 MB, while preserving high-fidelity rendering quality for spatial perception. Ultimately, this compact and physics-aligned representa-

tion bridges the gap between high-fidelity 3D reconstruction and constrained aerospace communication, paving the way for efficient teleoperation and autonomous visual navigation in future on-orbit servicing missions.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (NSFC) under Grant No. 62573351, 62022067, 62403383. We also acknowledge the support from Smart Satellite Technology Co., Ltd. in providing the Smart Satellite 1A mockup.

REFERENCES

- [1] N. Fallahiarezoodar and Z. H. Zhu, "Review of autonomous space robotic manipulators for on-orbit servicing and active debris removal," *Space: Science & Technology*, vol. 5, p. 0291, July 2025.
- [2] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: representing scenes as neural radiance fields for view synthesis," in *European Conference on Computer Vision (ECCV)*. Springer, 2020, pp. 405–421.
- [3] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Transactions on Graphics*, vol. 42, no. 4, pp. 1–14, July 2023.
- [4] S. Youn, S. Lee, G. Kim, W. Kwon, S.-H. Bae, and J. Oh, "Success-GS: survey of compactness and compression for efficient static and dynamic Gaussian splatting," Dec. 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2512.07197>
- [5] M. T. Bagdasarian, P. Knoll, Y. Li, F. Barthel, A. Hilsmann, P. Eisert, and W. Morgenstern, "3DGS.zip: a survey on 3D Gaussian splatting compression methods," *Computer Graphics Forum*, vol. 44, no. 2, Apr. 2025.
- [6] S. Jiang, C. Wang, X. Ren, L. Jiang, and Z. Wang, "A spacecraft 3D reconstruction method based on component-level energy accumulation of isar image sequences," *IEEE Transactions on Aerospace and Electronic Systems*, pp. 1–20, Mar. 2026.
- [7] P. Tang, Z. Li, Y. Chen, D. Sun, F. Wang, and Y.-Q. Jin, "G2-SDF: geometry-guided neural signed distance field for space target reconstruction," *IEEE Transactions on Aerospace and Electronic Systems*, pp. 1–15, Apr. 2026.
- [8] R. Mascaro and M. Chli, "Scene representations for robotic spatial perception," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 8, no. 1, pp. 351–377, May 2025.
- [9] Y. Liu, Z. Sun, L. Zhang, L. Xi, W. Dong, and F. Deng, "Spacecraft-NeRF: high-fidelity reconstruction of spacecraft by neural radiance field based implicit representation," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 6, pp. 15 182–15 194, Nov. 2025.
- [10] B. Han, C. Wang, X. Zhang, Z. Zhao, Z. Zhai, J. Liu, N. Liu, and X. Chen, "Pose estimation and neural implicit reconstruction toward noncooperative spacecraft without offline prior information," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 2, pp. 2612–2630, Apr. 2025.
- [11] D. Yang, Y. Zhang, G. Yu, J. Jiao, P. Huang, and B. Wang, "NeRF-based simultaneous pose estimation and 3D reconstruction for non-cooperative space target," *Aerospace Science and Technology*, vol. 168, p. 111010, 2026.
- [12] Y. Liao, Y. Di, H. Zhou, K. Zhu, M. Lu, Q. Duan, and J. Liu, "A survey on neural radiance fields," *ACM Computing Surveys*, vol. 58, no. 2, pp. 1–33, Sept. 2025.
- [13] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Transactions on Graphics*, vol. 41, no. 4, pp. 1–15, July 2022.
- [14] L. Liu, J. Gu, K. Z. Lin, T.-S. Chua, and C. Theobalt, "Neural sparse voxel fields," in *Neural Information Processing Systems (NeurIPS)*. Curran Associates, Inc., 2020.
- [15] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan, "Mip-NeRF: a multiscale representation for anti-aliasing neural radiance fields," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 5835–5844.
- [16] Y. Li, Q. Hu, T. Liang, D. Li, and Z. Ouyang, "Active 2dgs for 3D reconstruction of space targets under orbital constraints," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 4, pp. 4971–4983, Apr. 2026.
- [17] J. Wang, M. Liu, X. Yang, X. Wei, L. Wang, and N. Wang, "Satellite-GS: enhanced 2D Gaussian splatting for robust satellite reconstruction," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, pp. 1–15, Mar. 2026.
- [18] M. S. Ali, C. Zhang, M. Cagnazzo, G. Valenzise, E. Tartaglione, and S.-H. Bae, "Compression in 3D Gaussian splatting: a survey of methods, trends, and future directions," *IEEE Transactions on Circuits and Systems for Video Technology*, p. 1, Mar. 2026.
- [19] T. H. Park and S. D'Amico, "Improved 3D Gaussian splatting of unknown spacecraft structure using space environment illumination knowledge," in *2025 IEEE International Conference on Space Robotics (iSpaRo)*. IEEE, Dec. 2025.
- [20] Z. Fan, K. Wang, K. Wen, Z. Zhu, D. Xu, and Z. Wang, "LightGaussian: unbounded 3D Gaussian compression with 15x reduction and 200+ fps," in *Neural Information Processing Systems (NeurIPS)*, vol. 37. Curran Associates, Inc., 2024, pp. 140 138–140 158.
- [21] Q. Yang, X. Dongye, H. Guo, D. Weng, and L. Luo, "MiniGS: efficient 3D Gaussian splatting with full factors weighted pruning for scene representation," *Multimedia Systems*, vol. 32, no. 1, p. 66, Jan. 2026.
- [22] J. C. Lee, D. Rho, X. Sun, J. H. Ko, and E. Park, "Compact 3D Gaussian representation for radiance field," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 21 719–21 728.
- [23] K. Navaneet, K. Pourahmadi Meibodi, S. Abbasi Koohpayegani, and H. Pirsiavash, "CompGS: smaller and faster Gaussian splatting with vector quantization," in *European Conference on Computer Vision (ECCV)*, vol. 15090. Springer, 2024, pp. 330–349.
- [24] S. Niedermayr, J. Stumpfegger, and R. Westermann, "Compressed 3D Gaussian splatting for accelerated novel view synthesis," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 10 349–10 358.
- [25] S. Girish, K. Gupta, and A. Shrivastava, "Eagles: efficient accelerated 3D Gaussians with lightweight encodings," in *European Conference on Computer Vision (ECCV)*, vol. 15121. Springer, 2025, pp. 54–71.
- [26] T. Lu, M. Yu, L. Xu, Y. Xiangli, L. Wang, D. Lin, and B. Dai, "Scaffold-GS: structured 3D Gaussians for view-adaptive rendering," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 20 654–20 664.
- [27] Y. Chen, Q. Wu, W. Lin, M. Harandi, and J. Cai, "Hac: hash-grid assisted context for 3D Gaussian splatting compression," in *European Conference on Computer Vision (ECCV)*. Springer, 2025, pp. 422–438.
- [28] Y. Chen, Q. Wu, W. Lin, M. Harandi, and J. Cai, "Hac++: towards 100x compression of 3D Gaussian splatting," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 11, pp. 10 210–10 226, 2025.
- [29] H. Xu, X. Wu, and X. Zhang, "Learning hierarchical sparse transform coding of 3DGS," 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2505.22908>
- [30] Y. Wang, Z. Li, L. Guo, W. Yang, A. C. Kot, and B. Wen, "ContextGS : compact 3D Gaussian splatting with anchor level context model," in *Neural Information Processing Systems (NeurIPS)*, vol. 37. Curran Associates, Inc., 2024, pp. 51 532–51 551.
- [31] Z. Liu, R. Song, Y. Huang, Y. Hu, X. Zhang, J. Shao, Z. Lin, and J. Zhang, "Feed-forward 3D Gaussian splatting compression with long-context modeling," 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2512.00877>
- [32] B. Huang, Z. Yu, A. Chen, A. Geiger, and S. Gao, "2D Gaussian splatting for geometrically accurate radiance fields," in *ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)*. Association for Computing Machinery, 2024.
- [33] A. Guédon and V. Lepetit, "Sugar: surface-aligned Gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 5354–5363.
- [34] X. Jin, R. Jin, B. Li, D. Zou, and W. Yu, "PlanarGS: high-fidelity indoor 3D Gaussian splatting guided by vision-language planar priors," in *Neural Information Processing Systems (NeurIPS)*, 2025.
- [35] S. Liu, T. Deng, H. Zhou, L. Li, H. Wang, D. Wang, and M. Li, "Mg-SLAM: structure Gaussian splatting SLAM with Manhattan world hypothesis," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 17 034–17 049, 2025.
- [36] S. Zhou, H. Chang, S. Jiang, Z. Fan, Z. Zhu, D. Xu, P. Chari, S. You, Z. Wang, and A. Kadambi, "Feature 3DGS: supercharging 3D Gaussian splatting to enable distilled feature fields," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 21 676–21 685.

- [37] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister, “LangSplat: 3D language Gaussian splatting,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 20 051–20 060.
- [38] J. Cen, J. Fang, C. Yang, L. Xie, X. Zhang, W. Shen, and Q. Tian, “Segment any 3D Gaussians,” in *The Association for the Advancement of Artificial Intelligence (AAAI)*, vol. 39, no. 2, Apr. 2025, pp. 1971–1979.
- [39] D. Cai, J. Heikkilä, and E. Rahtu, “GS-Pose: generalizable segmentation-based 6D object pose estimation with 3D Gaussian splatting,” in *2025 International Conference on 3D Vision (3DV)*, 2025, pp. 1001–1011.
- [40] B. Jiang, Y. Xie, J. Li, N. Li, B. Chen, and S.-T. Xia, “IGSPAD: inverting 3D Gaussian splatting for pose-agnostic anomaly detection,” in *Proceedings of the 32nd ACM International Conference on Multimedia (MM)*. Association for Computing Machinery, 2024, pp. 10 229–10 237.
- [41] B. Matteo, T. Tsesmelis, S. James, F. Poiesi, and A. Del Bue, “6DGS: 6D pose estimation from a single image and a 3D Gaussian splatting model,” in *European Conference on Computer Vision (ECCV)*. Springer, 2024, pp. 420–436.
- [42] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar, and R. Girshick, “Segment anything,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2023, pp. 3992–4003.
- [43] R. B. Rusu, N. Blodow, and M. Beetz, “Fast point feature histograms (FPFH) for 3D registration,” in *2009 IEEE International Conference on Robotics and Automation (ICRA)*, May 2009, pp. 3212–3217.
- [44] Y. Jiang, J. Tu, Y. Liu, X. Gao, X. Long, W. Wang, and Y. Ma, “GaussianShader: 3D Gaussian splatting with shading functions for reflective surfaces,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 5322–5332.
- [45] P. Hedman, J. Philip, T. Price, J.-M. Frahm, G. Drettakis, and G. Brostow, “Deep blending for free-viewpoint image-based rendering,” vol. 37, no. 6, Dec. 2018.
- [46] National Aeronautics and Space Administration, “NASA 3D resources — 3D models.” [Online]. Available: <https://nasa3d.arc.nasa.gov/models>
- [47] European Space Agency, “ESA science satellite fleet.” [Online]. Available: <https://scifleet.esa.int>
- [48] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [49] S. Li, C. Wu, H. Li, X. Gao, Y. Liao, and L. Yu, “GSCodec Studio: A modular framework for Gaussian splat compression,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 4, pp. 5483–5496, 2026.